ICT305 Topics in IT

Web3 and VR/AR

CIHE Australia

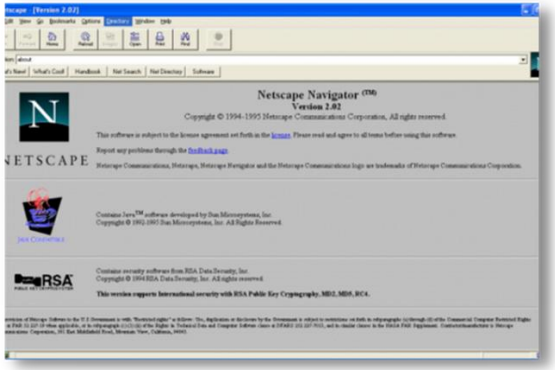
Agenda

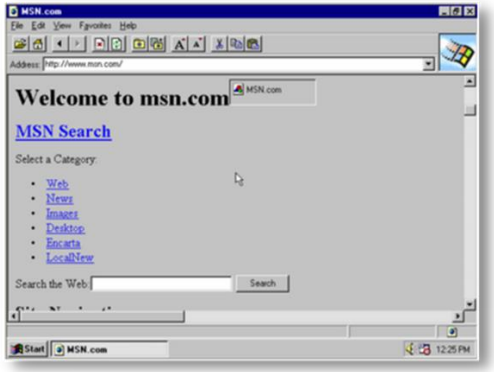
- Web3 History
- Web3 Architecture
- Web3 Examples
- Virtual reality (VR) and augmented reality (AR)
- Elements in VR/AR system
- Applications of AR/VR
- Future of AR and VR to business

Web3

Web1 ("Static Web – Read Only")

- **Web1** was the first version of the Internet, primarily consisting of simple web pages that were manually updated and provided basic information to users.
- It was mainly used for displaying and reading information, but interaction was limited.





Web2 (“Dynamic Web”)

- **The second** and **current version** of the Internet introduced new technologies and features that permitted greater user interaction and collaboration.
- **Platforms** such as social media, blogs, and wikis, allowed users to create, share, and comment on content, facilitating connections between users.
- This led to a more social and interactive web experience, by allowing users to create the content they wanted to see.

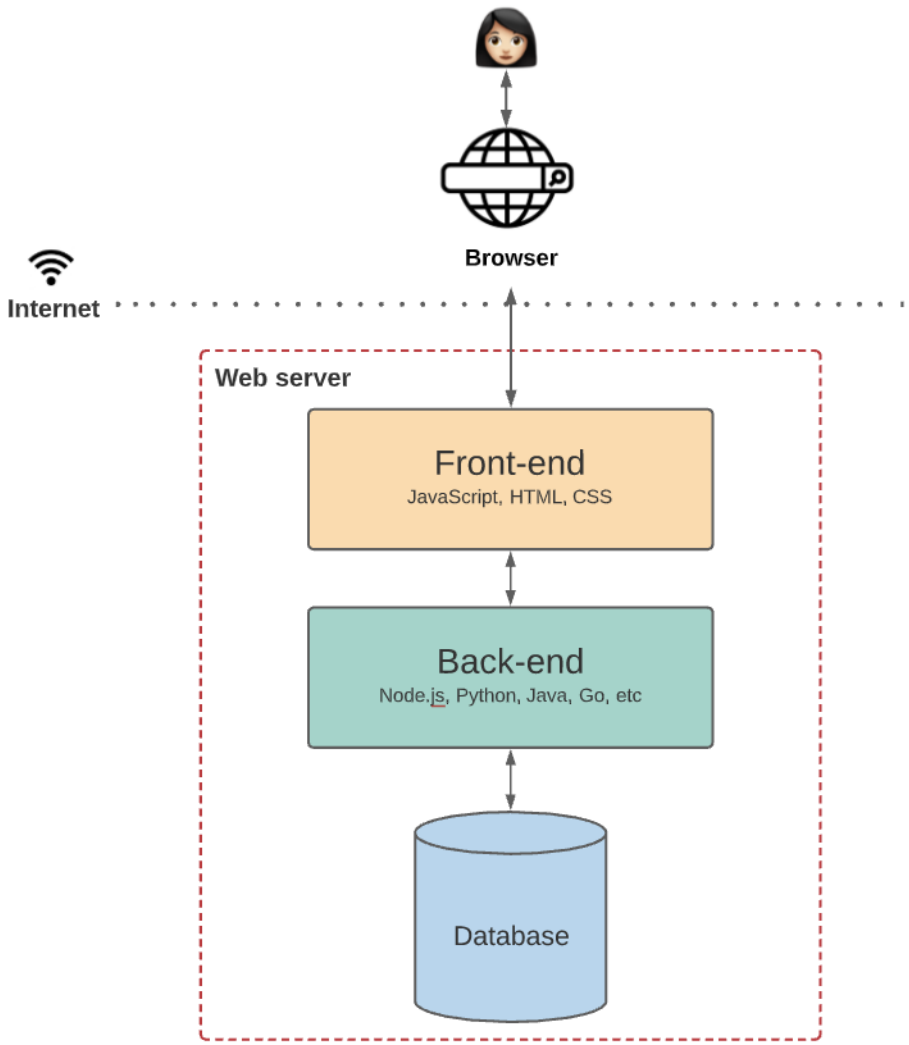
Web3 ("Symantec Web")

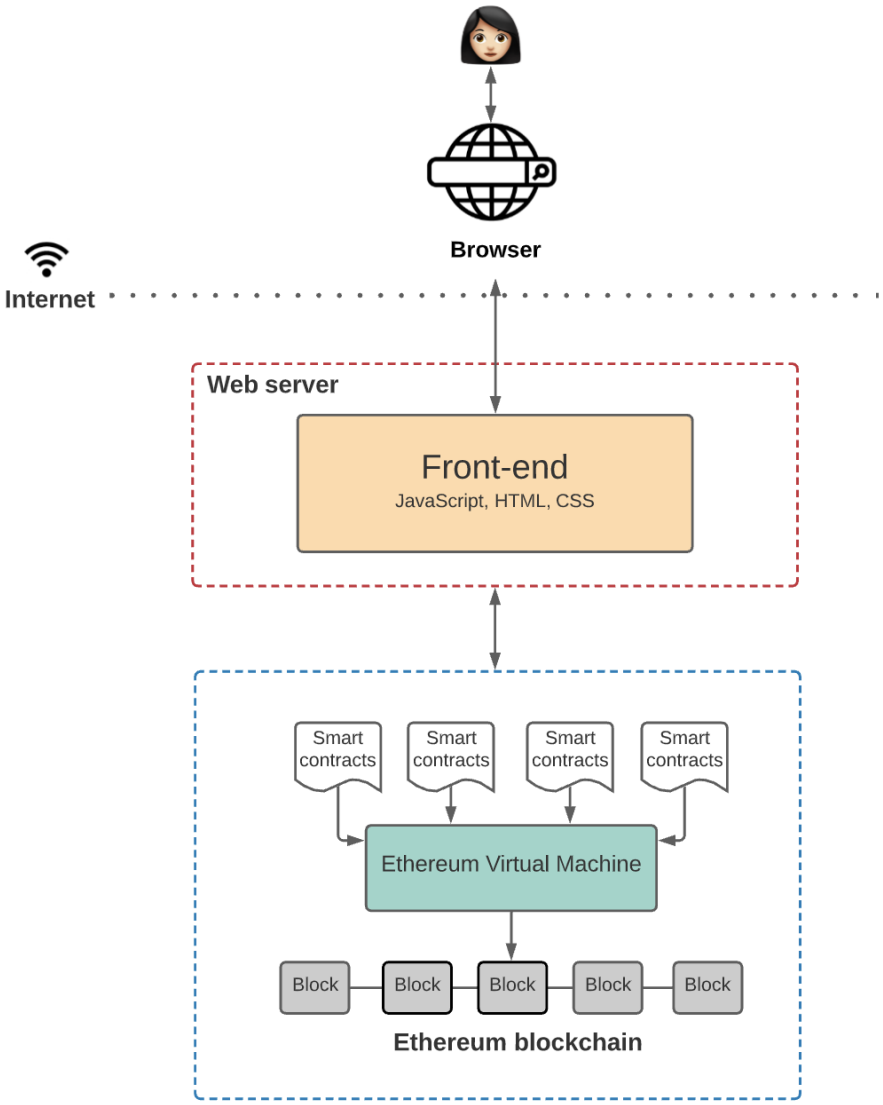
- **Web3** aims to make the Internet more intelligent, connected, and user-centric, by using new technologies such as blockchain, artificial intelligence, and natural language processing.
- The goal is to create a ***more decentralised and equitable Internet***, where individuals have more control over their data and online identity.
- It also aims to make it easier for machines and software to understand and interact with information on the Internet.
- **Web3** is currently a work in progress and has not yet been *precisely defined*.
- However, the main principle is for it to be decentralised - rather than being controlled by governments and corporations, as is the case with previous iterations of the web.

Web3 Applications

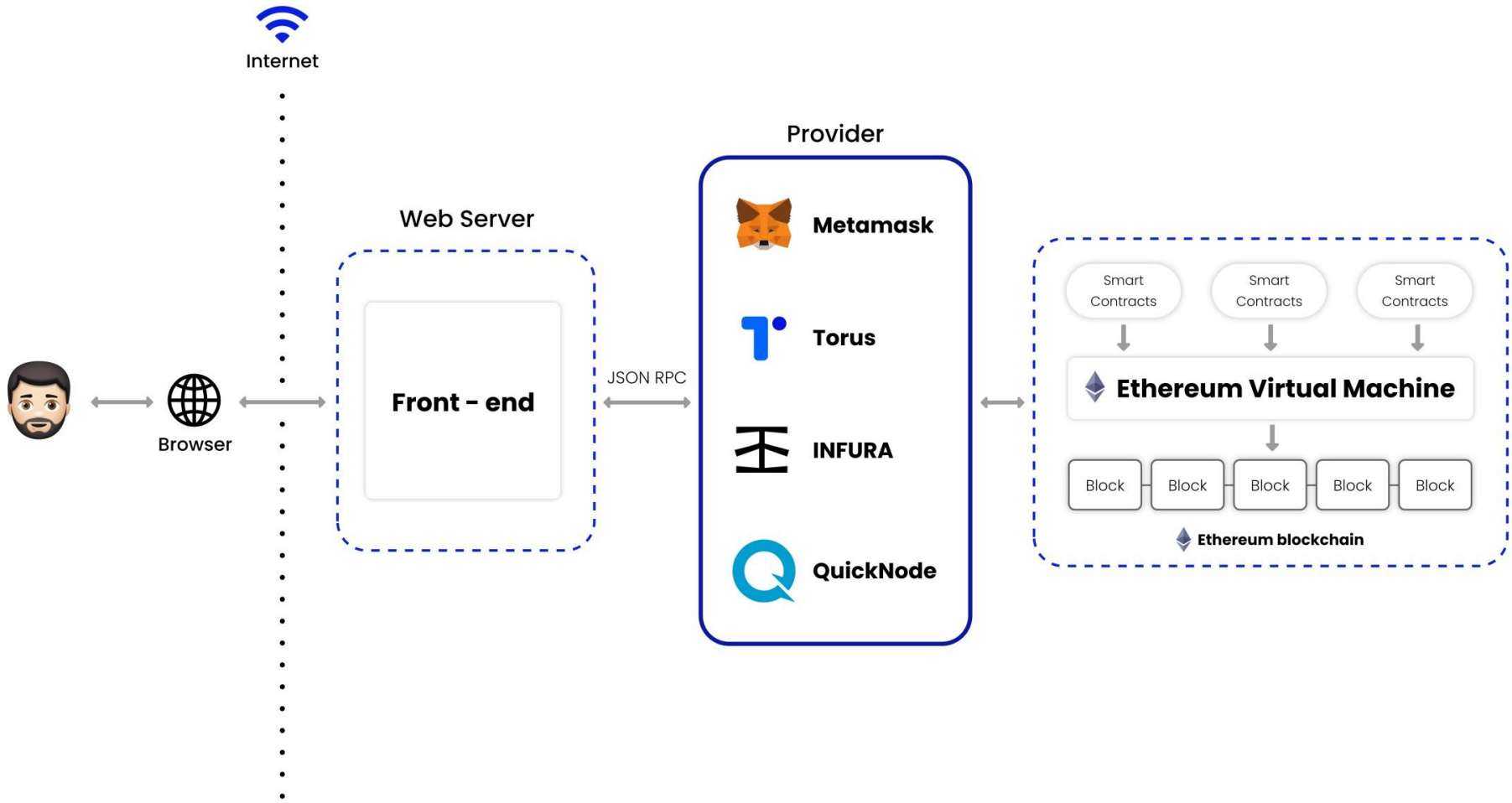
- **Web3 applications** incorporate AI and machine learning technology.
  - Most of the Web3 Dapps (Decentralised Apps) that are live today involve cryptocurrency and finance.
  - In the future, all kinds of smarter and more user-centric apps will be created.
- **An example** of an app employing Web3 technology is **Siri**.
  - Apple's AI assistant lets users control their surroundings and devices with voice commands.
- **Another popular Web3 app** is the browser **Brave**, which connects users with DApps, crypto wallets and other Web3 technologies.
  - Without addons, Web3 does not yet work on traditional browsers such as Chrome, Firefox, Safari or Microsoft Edge.
  - This could change with the arrival of browsers that natively support Web3, such as Opera and Brave.

Web2 vs Web3 Architecture





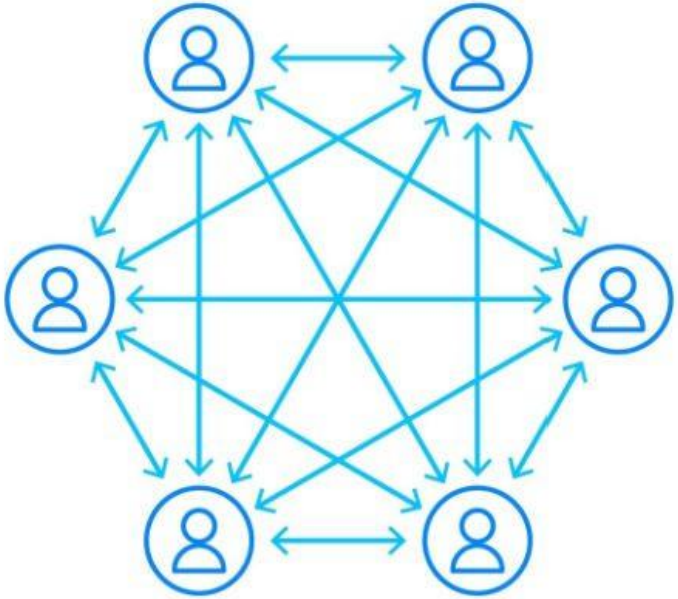
Web3 Architecture



Web3: Decentralization

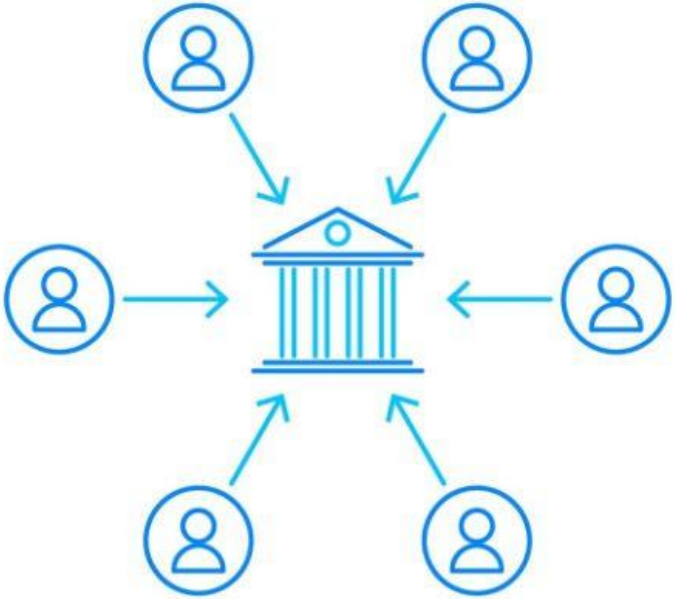
- Today, the online infrastructures used by popular sites and services are usually owned by corporations, and so to a certain extent, are controlled by government regulations.
- **Blockchain**, as a new method of storing data online, **changes this**.
- **Blockchain** is built around the *concepts of encryption and distributed computing*:
  - refers to the use of multiple devices working together to **maintain** and **validate** a decentralised and secure digital ledger.

Web3: Decentralization



DeFi

Decentralized Finance



CeFi

Centralized Finance

Blockchain

- It can be difficult to wrap your mind around how blockchain works.
- It is best described as **a digital ledger** that records transactions in a secure and tamper-proof way, using a decentralised network of users.
- It acts as a digital book, where *transactions are recorded in blocks and linked together*, making it almost impossible to tamper with or change recorded transactions.

Blockchain

A new transaction is entered.

The transaction is then transmitted to a network of peer-to-peer computers scattered across the world.

This network of computers then solves equations to confirm the validity of the transaction.

Once confirmed to be legitimate transactions, they are clustered together into blocks.

These blocks are then chained together creating a long history of all transactions that are permanent.

The transaction is complete.

Risks/Downsides of Web3

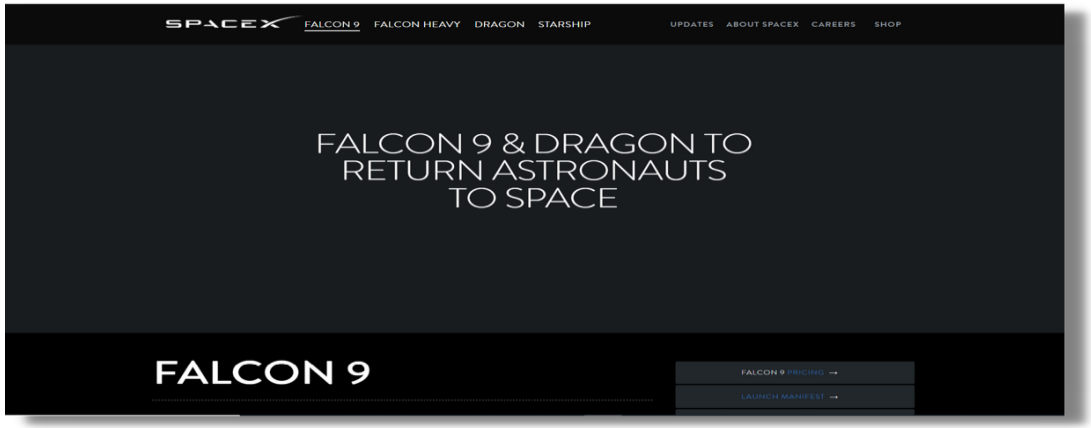
- The lack of centralised gatekeepers in Web3 could pose a significant risk to users.
- While Web3's decentralised ownership is seen to empower individuals, the lack of oversight increases consumer risk, as was seen in the collapse of the major cryptocurrency exchange FTX.
- *Steps are being taken to mitigate Web3's risks*, but two main concerns are:
  - **Regulation**: Web3's decentralised nature makes regulation virtually impossible.
    - For example, a fraudulent website, or one that infringes copyright on a decentralised blockchain, will be difficult (if not impossible) to remove, since it is hosted on a peer-to-peer network where the content is stored across multiple nodes/locations.
  - **Privacy**: Personal data may be collected and used for targeted advertising and other purposes without the individual's consent

Web 3 Use Examples

- **Web3** can seem like an abstract concept, as most will not have made visible use of Dapps (e.g., tiny racing) or Web3 websites.
- To access the sites below you will need a browser that supports them, such as Opera (www.opera.com).
- Here are some examples:
  - **DTube** - ipfs://dtube.crypto
  - **DTube** is the Web3 version of YouTube.
  - **YouTube** has terms and conditions that regulate the content that is posted and what content can be removed.
  - **DTube**, however, is not regulated, therefore an uploaded video is not monitored, and cannot be taken down or edited.
  - An advantage of **DTube** is that you can watch videos without advertisements, while a major disadvantage is that inappropriate materials which could be accessed by younger users cannot be removed.

Web 3 Use Examples

- SpaceX - ipfs://spacex.eth/index.html
- We could not work out whether the SpaceX website is official. If it’s legitimate, it needs a lot of work.
- If it’s fake then Elon has some work to do to take it down.



Web3 Examples

- **Ethereum Web3** - ipfs://ethereum.eth
  - This the official Web3 page of the cryptocurrency Ethereum, using its Web3 domain extension .eth.
  - Web3 Domain extensions such as .eth and .crypto are not regulated by ICANN or any other regulatory body.
  - This means that there is no dispute resolution service, so the only resolution to a Web3 domain dispute is to purchase the domain from the registrant, if they are contactable.
  - **Squatters** list their infringing Web3 domains on the auction website opensea.io.

Next Agenda

VR/AR

1. Definition of virtual reality (VR) and augmented reality (AR)
2. Applications of AR/VR
3. VR and AR in education industry
4. Elements in VR/AR system
5. Challenges to AR and VR
6. Future of AR and VR to business

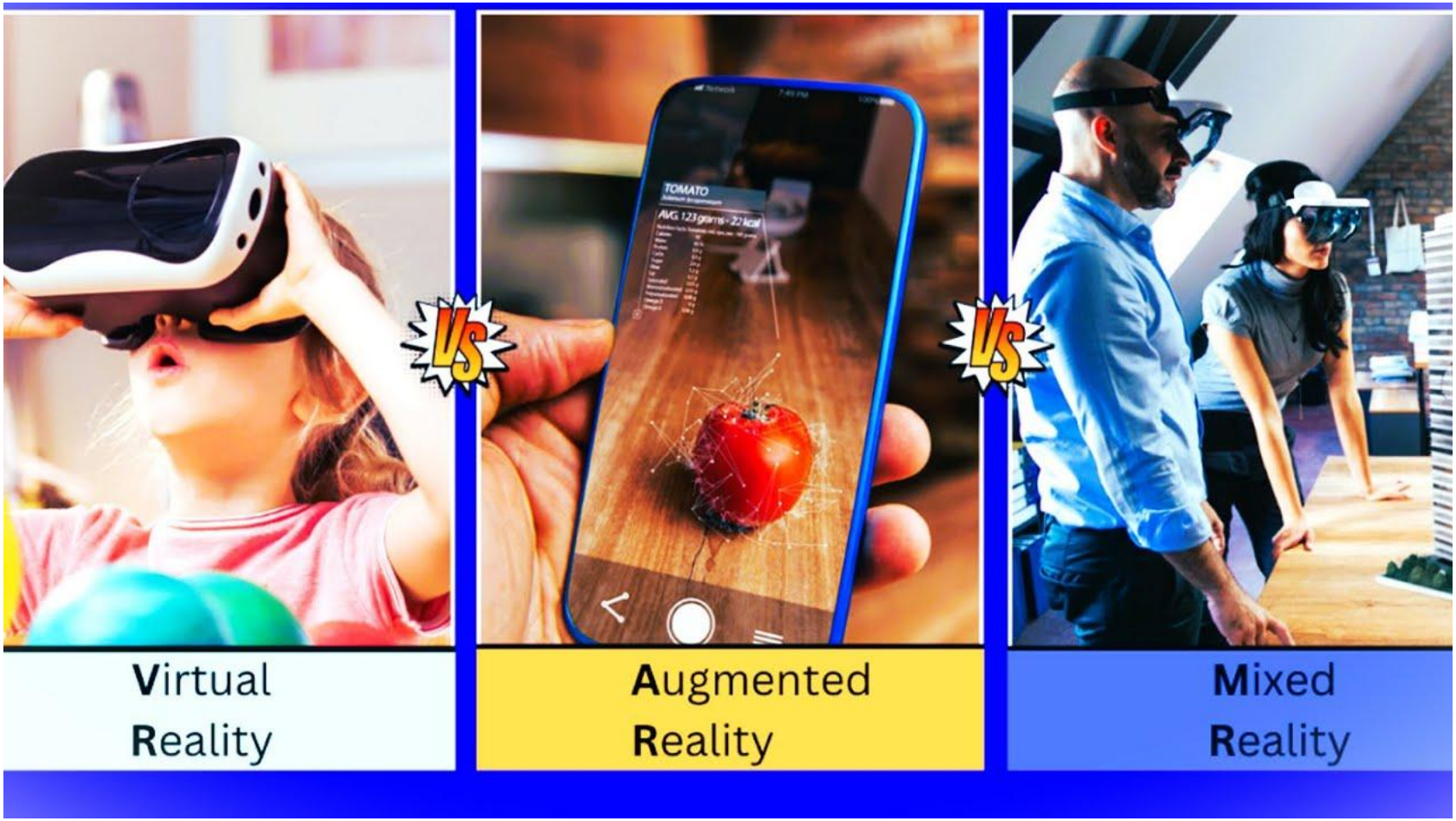
Definition of virtual reality (VR) and augmented reality (AR)

- **Virtual reality (VR)** is a technology where
  - a VR device creates a new reality from scratch and brings the device user to the reality while almost cutting him/her from the real world.
- **Augmented reality (AR)** is a technology where
  - an AR device, which can be in the form of our regular smartphones, creates a new reality from materials taken from the real world, which are usually taken using a phone camera.

Difference between VR and AR

- **VR** almost cuts the user from the real world due to the use of a pair of headsets and the VR device itself that completely blocks the user's vision.
- **AR** does not require the user to put a pair of headsets and the device, which is usually a smartphone or tablet, thus not completely blocking the user's vision at all.

Rise of reality technologies



VR is better than AR?

- **VR** creates a completely different reality for the user while **AR** only does half of the job or even less.
- So, it is safe to assume that **VR** is much better than **AR** in terms of creating a new reality and bringing the user to it.
- However, if the user does not want to really separate him/herself from the real world due to certain reasons, then AR is the more suitable option for him/her.
- So, the decision to use VR or AR depends on the interests and needs of the user, but still, in terms of technology, VR is absolutely better than AR.

Some applications of AR and VR

- Both VR and AR appear due to the demands of making a digital reality for many purposes. So, it should not be too hard to find the applications of AR and VR in real life.
- One of the applications of AR that we can find in our daily life is photo or video filter.
  - By using photo or video filters, we have altered or augmented the reality that has been take by our smartphone camera, so that the result of the filtered photo or video deserves to be called an AR.
  - Such filters are now quite common in our society because anyone who accesses media social can use the filters.
- There are many other significant applications of AR and VR besides photo or video filters.
  - Those applications include the applications in video games, medical exams, engineering, such as simulation of a building construction, project explanation, and so on.

Some applications of VR and AR in education industry

VR and AR in education industry

- If we use AR technology in the microbiology class, we can see the cells that are being explained in more details in 3D.
  - We can also see many kinds of cells from various organisms, and we do not have to go through the trouble of preparing the cells and microscope for that.
- **VR** can even give a better vision to students than what AR can give regarding the materials being studied.
  - However, it seems that only high-budget schools or colleges that have applied VR in education due to the relatively expensive price of a VR device.
- This is only one example of VR and AR in education, so that there are many other examples out there:
  - the use in explaining our solar system in an astronomy class,
  - the use in explaining atoms in chemistry class, and so on.

Top 5 Elements of VR

1: Viewing System

2: Tracking System

3: Interactivity Element

4. Artistic Inclination

5: Sensory Management System

Elements in AR system

- Digital and physical worlds
- Interactions made in real time
- Accurate 3D identification of virtual and real objects.
- 2 types of AR systems are used dominantly:
  - **Marker-based AR** is created using image recognition to identify objects already programmed into your AR device or application
  - **Marker-less AR** is more complex as there's no point in which your device will focus on. Thus, your device must recognize items as they appear in view

Marker-based AR

10 minute in-class mini-exercise

Create simple AR using your mobile app: Arloopa

Google’s AR and VR

- Google has launched its new product called Google’s AR and VR. AR and VR create new reality to help and ease our life with their implementation. So, Google uses the new technologies to help bridge digital world and physical world.
- Google AR and VR utilizes the camera of the user’s smartphone and provide the user with information they need, such as the name of the street, navigation, simulation, and so on.
- So, by using Google’s AR and VR, its user will be able to obtain information that can be very helpful for them.

Challenges to AR and VR

- **Limited user-experience**: most of the available devices are incompetent in providing full-fledged AR/VR experiences
- **Limited use cases**: consumers see no purpose in buying them, except for the techies and gamers
- **Aesthetics & comfort**: aesthetics are a mainstream subject looked at by consumers. In short, device aesthetics are a major aspect that triggers the adoption. Currently, AR/VR wearables come with very bulky form factors.
- **Digital fatigue**: television sets and mobile phones are loved by consumers, but partaking in virtual experience is simply exhausting for consumers.
- **High price on new tech**: AR/VR devices are greatly expensive for modern users. Besides, wearable and headset creation is also expensive and labor-intensive.

Conclusions

- Future of AR and VR in business